7. Write a C program for the following ciphertext was generated using a simplesubstitutionalgorithm.53‡‡†305))6*;4826)4‡.)4‡);806*;48†8¶60))85;;]8*;:‡*8†83(88)5*†;46(;88*96*?;8)*‡(;485);5*†2:*‡(;4956*2(5*—4)8¶8*;4069285);)6†8)4‡‡;1(‡9;48081;8:8‡1;48†85;4)485†528806*81(‡9;48;(88;4(‡?34;48)4‡;161;:188;‡?;

Aim:

To decrypt a message encrypted using a simple substitution cipher.

Procedure:

1. Read the ciphertext.
2. Count the frequency of each ciphertext letter.
3. Use common English letter frequencies to estimate substitutions.
4. Replace ciphertext letters with their estimated plaintext letters.
5. Display the resulting text.

Code:

```
#include <stdio.h>
#include <ctype.h>
int main() {
   char text[1000];
   char freq[] = "ETAOINSHRDLCUMWFGYPBVKJXQZ";
   int count[26] = {0}, map[26], i, j;
   printf("Enter ciphertext: ");
   fgets(text, sizeof(text), stdin);
   for (i = 0; text[i]; i++)
      if (isalpha((unsigned char)text[i]))
```

```
        count[toupper(text[i]) - 'A']++;
    for (i = 0; i < 26; i++)
        map[i] = -1;

    for (i = 0; i < 26; i++) {
        int max = -1, pos = -1;
        for (j = 0; j < 26; j++)
            if (count[j] > max) {
                max = count[j];
                pos = j;
            }
        if (pos >= 0 && max > 0) {
            map[pos] = freq[i] - 'A';
            count[pos] = -1;
        }
    }
    for (i = 0; text[i]; i++) {
        if (isalpha((unsigned char)text[i])) {
            int x = toupper(text[i]) - 'A';
            if (map[x] >= 0)
                text[i] = map[x] + 'A';
        }
    }
    printf("Estimated plaintext:\n%s", text);
    return 0;
}
```

Input:

Enter ciphertext: KHOOR

Output:

